

Project Proposal

Description

People on the Mass Open Cloud (MOC) Alliance want to deploy Open Claw Agents on the New England Research Cloud (NERC) OpenShift Clusters. However, there are some concerns regarding the dangers of running Open Claw agents on the NERC clusters. What if it leaks secrets? What if a user prompts the agent to delete system critical files? There are a plethora of problems in addition to these, and so, a new question is formulated: Is it even worth creating the infrastructure to securely support Open Claw, or other LLM-driven agents, on the NERC?

The goal of this project is to conduct a cost-benefit analysis to recommend whether OpenClaw Agents should be deployed on the NERC, and what needs to be implemented to successfully guardrail these agents. For the scope of this project, nothing will be hosted on local hardware, Sally O'Malley's claw-installer will be examined, and OpenClaw will be tested using a cloud provider, local provider, and both.

There are a few key areas that need to be addressed: resource access, secret keeping, and input/output screening. Resource access has been the biggest deterrent preventing users from locally running OpenClaw, and I plan to resolve this issue through OpenShift RBAC and ANP. Another important security flaw is in secret keeping. Snyk found that 7% of the OpenClaw skills have critical security vulnerabilities when it comes to protecting sensitive credentials. I plan to patch this issue through the use of Hashicorp Vault, in order to safely access and store sensitive secrets, such as primarily the API key for the LLM provider. The last major security flaw with OpenClaw is prompt injection. A way to fix this is to screen user input. The TrustyAI Guardrails Orchestrator is a tool that accomplishes this. After implementing these tools, I will do a cost benefit analysis to determine if the benefits are greater than the tradeoffs.

Tradeoffs

The first problem I anticipate having is in the setup. Thus far, no one has deployed an OpenClaw agent on the MOC, so it will be a first time for everything. Another tradeoff is that due to the addition of TrustyAI, there will be more latency as both input and output will be put through detector models. Additionally, because there are a number of tools being involved to implement these guardrails, it will require much more maintenance in order to keep everything up to date. Another tradeoff is that there will be multiple use cases for OpenClaw, so certain policies may not work for everyone. Finally, due to the nature of the tools being used, there will be considerably more heavy resource/compute overhead.

Deliverables

Throughout this entire process, I will be documenting everything I am doing. At the halfway point, I will do a demo of the first deployed OpenClaw agent with at least one of my aforementioned guardrails. At the end, I will have formal documentation (report or Ops) with a recommendation, a cost-benefit analysis, and a video demo of the fully guardrailed agent.

